

NANYANG JUNIOR COLLEGE  
JC 2 PRELIMINARY EXAMINATIONS  
Higher 2

CANDIDATE  
NAME

CLASS

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**BIOLOGY**

Paper 1 Multiple Choice

**9648/01**

**28 September 2016**

**1 hour 15 min**

Additional Materials: Multiple Choice Answer Sheet

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**READ THESE INSTRUCTIONS FIRST**

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Write your name and CT on the Answer Sheet in the spaces provided unless this has been done for you.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

**Read the instructions on the Answer Sheet very carefully.**

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

Calculators may be used.

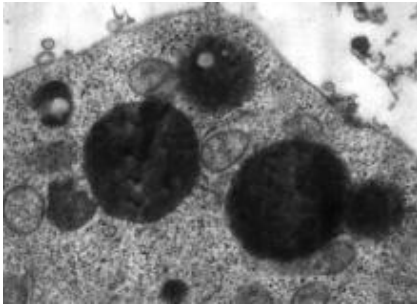
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This document consists of **27** printed pages and **1** blank page.

**[Turn over**

- 1 In an active cell, the pathway of a protein synthesis to maturation involves the endomembrane system.

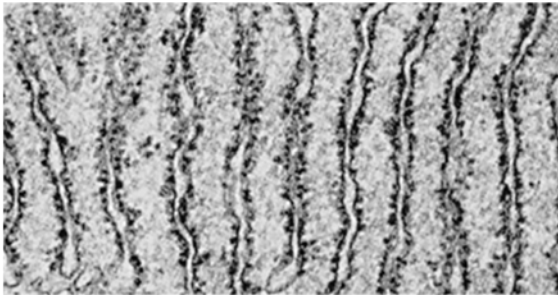
(I)



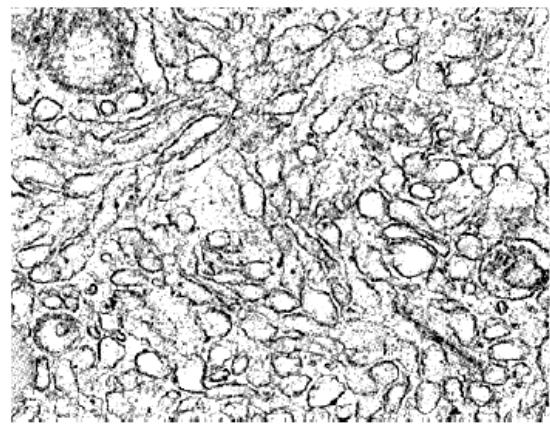
(II)



(III)



(IV)



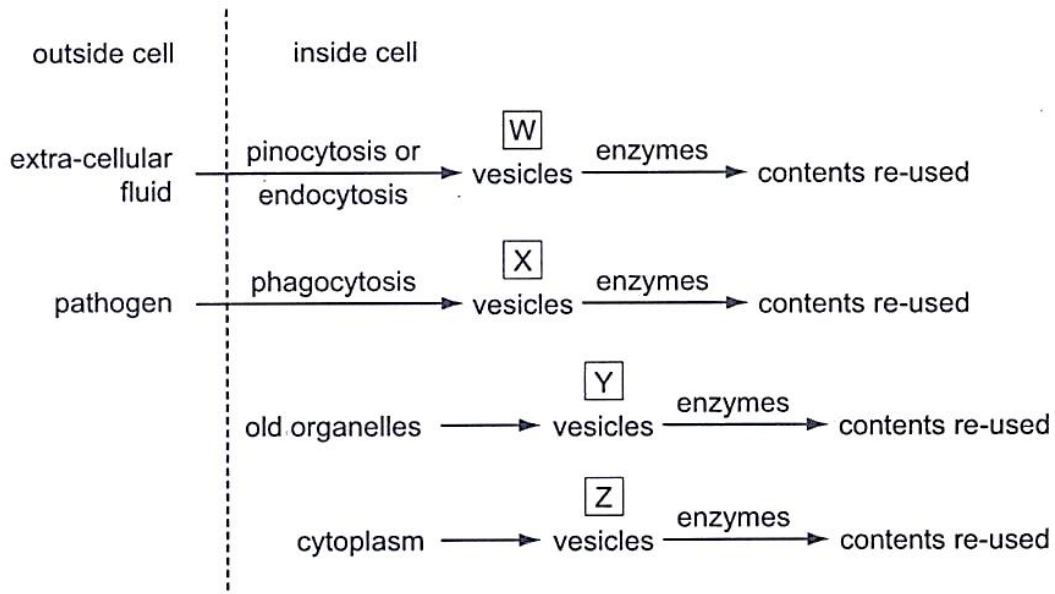
The following are several processes that occur in an active cell.

- w. modification of carbohydrates on protein
- x. folding of protein into its 3 dimensional conformation
- y. synthesis of steroids
- z. storage of synthesized proteins

Where in the above diagrams do these processes take place?

|   | (I) | (II) | (III) | (IV) |
|---|-----|------|-------|------|
| A | z   | w    | y     | x    |
| B | z   | w    | x     | y    |
| C | y   | w    | x     | z    |
| D | z   | y    | x     | w    |

2 The flow chart shows processes which takes place inside animal cells.



Which processes require the activity of lysosomes?

- A W, X, Y and Z
- B W and X only
- C X and Y only
- D Y and Z only

- 3 Which of the following lists of processes matches the type of movement across membrane correctly?

|          | <b>Passive diffusion</b>                  | <b>Facilitated diffusion</b>                        | <b>Active transport</b>                     | <b>Endocytosis</b>       | <b>Exocytosis</b>                     |
|----------|-------------------------------------------|-----------------------------------------------------|---------------------------------------------|--------------------------|---------------------------------------|
| <b>A</b> | Exit of CO <sub>2</sub> from mitochondria | Chemiosmosis of H <sup>+</sup> through ATP synthase | Exit of Ca <sup>2+</sup> from synaptic knob | Entry of HIV             | Budding of influenza virus            |
| <b>B</b> | Exit of CO <sub>2</sub> from mitochondria | Exit of K <sup>+</sup> from neuron                  | Entry of Na <sup>+</sup> into neuron        | Entry of HIV             | Budding of influenza virus            |
| <b>C</b> | Entry of O <sub>2</sub> into mitochondria | Exit of K <sup>+</sup> from neuron                  | Exit of Ca <sup>2+</sup> from synaptic knob | Entry of influenza virus | Release of insulin from $\beta$ cells |
| <b>D</b> | Entry of O <sub>2</sub> into mitochondria | Chemiosmosis of H <sup>+</sup> through ATP synthase | Entry of Na <sup>+</sup> into neuron        | Entry of influenza virus | Release of insulin from $\beta$ cells |

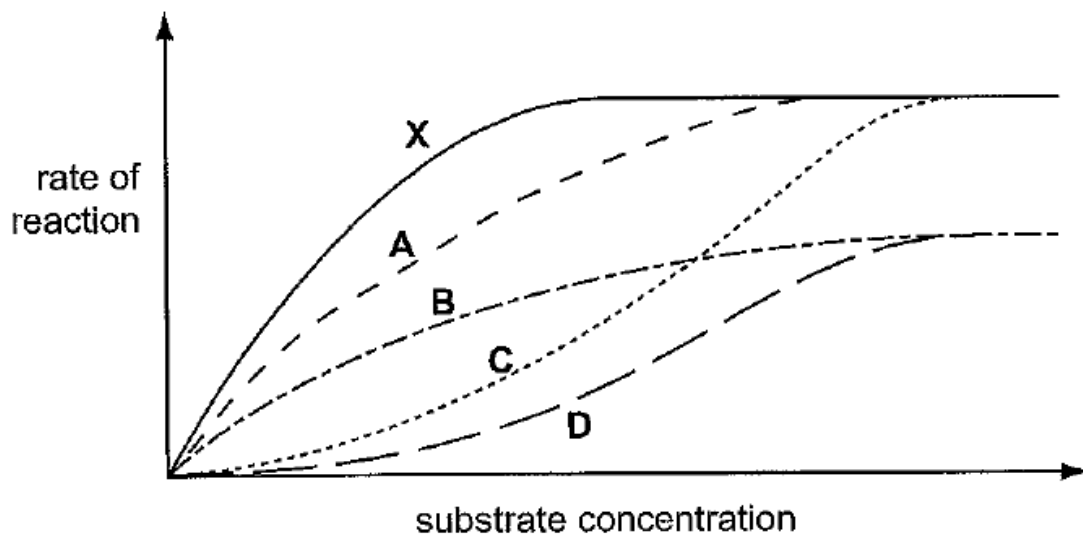
- 4 Raffinose is a trisaccharide which can be degraded by enzymes. The results of two different enzymatic incubations are shown in the table below:

| <b>Enzyme used</b> | <b>Products</b>        |
|--------------------|------------------------|
| Sucrase            | Melibiose and fructose |
| Galactosidase      | Galactose and sucrose  |

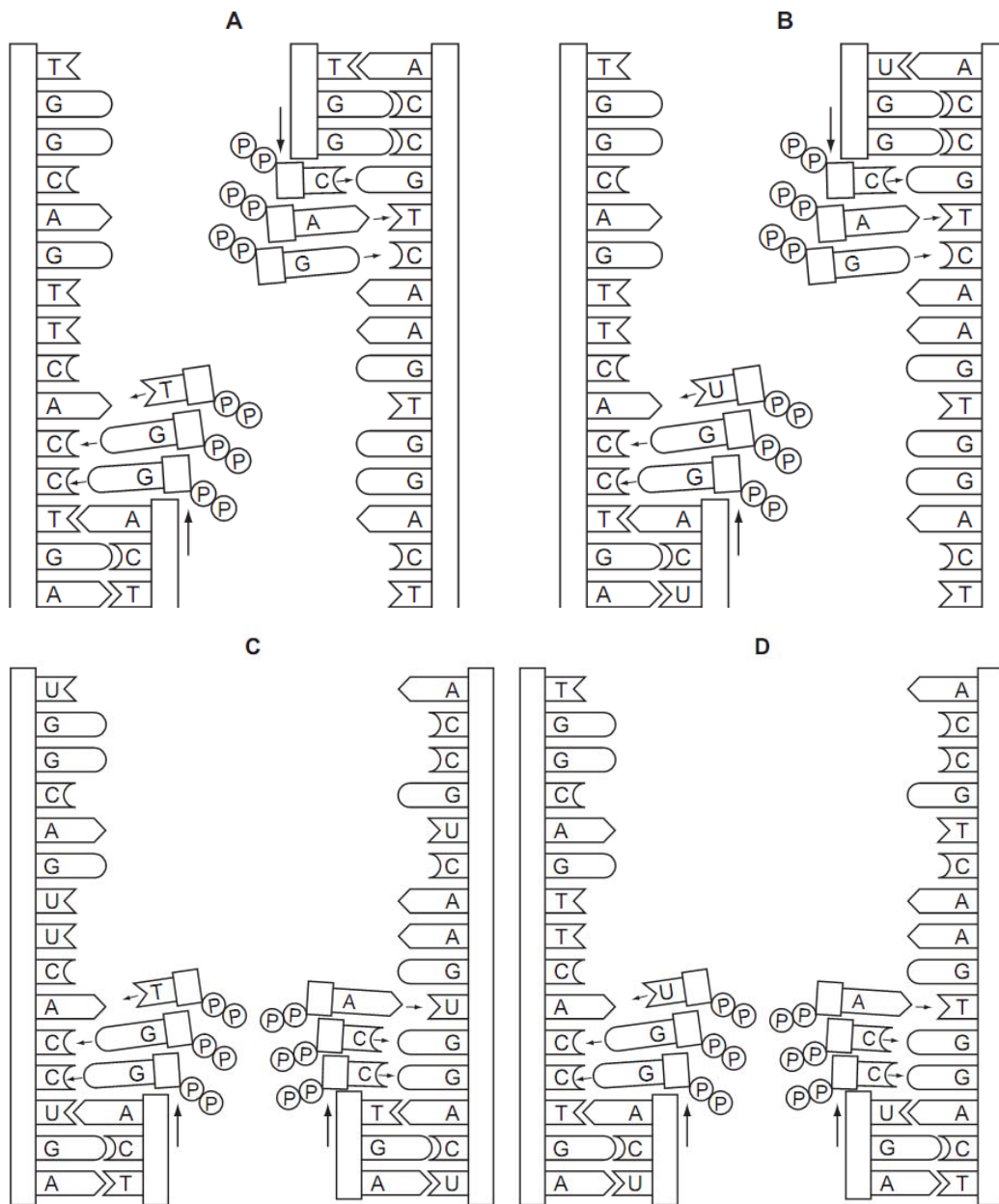
Which of following statements are consistent with the results shown above?

- I Raffinose is composed of three different monosaccharides.
  - II Melibiose is a disaccharide.
  - III One of the products of acid hydrolysis of raffinose is glucose.
  - IV The products of raffinose digestion by sucrase and galactosidase respectively will yield a brick-red precipitate when heated with Benedict's reagent.
- A** I and III only  
**B** II and IV only  
**C** I, II and III only  
**D** All of the above

- 5 The reaction rate of salivary amylase on starch decreases as the concentration of chloride ions is reduced. Which of the following describes the role of the chloride ions?
- A Allosteric inhibitors
  - B Co-enzymes
  - C Co-factors
  - D Competitive inhibitors
- 6 In the graph, X represents the initial rate of reaction of an enzyme in increasing substrate concentrations under optimum temperature and pH, in the absence of an inhibitor. Which curve represents the result when the same experiment is carried out under a lower temperature?



- 7 Which diagram shows the semi-conservative replication of a section of a molecule of DNA?



- 8 The diagram represents part of a DNA molecule.

|                                |
|--------------------------------|
| G A T A C C A<br>C T A T G G T |
|--------------------------------|

| Mutation                            | Name         |
|-------------------------------------|--------------|
| from purine to other purine         | transition   |
| from pyrimidine to other pyrimidine | transition   |
| from purine to pyrimidine           | transversion |
| from pyrimidine to purine           | transversion |

Which diagram shows the DNA molecule with only transversion(s)?

**A**

|                                |
|--------------------------------|
| G T T A T C A<br>C A A T A G T |
|--------------------------------|

**B**

|                                |
|--------------------------------|
| G A A A C A A<br>C T T T G T T |
|--------------------------------|

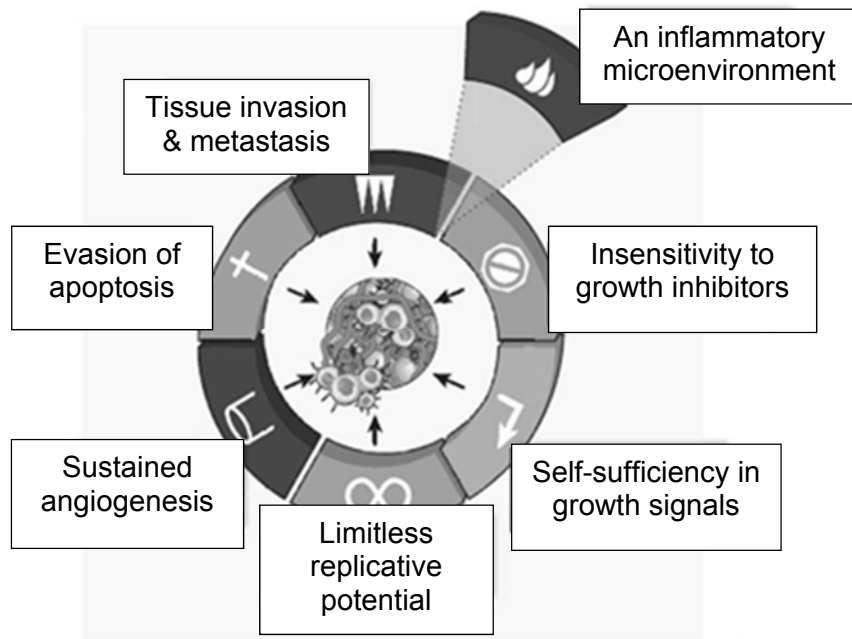
**C**

|                                |
|--------------------------------|
| A A T A C C A<br>T T A T G G T |
|--------------------------------|

**D**

|                                |
|--------------------------------|
| G A T A T C A<br>C T A T A G T |
|--------------------------------|

9 The diagram illustrates the hallmarks of cancer.



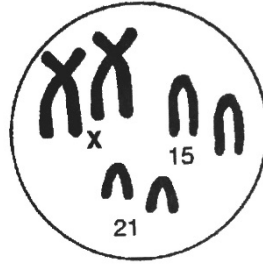
Which of the following statements correctly describe the changes in cancer cells?

|     |                                                                                                                  |
|-----|------------------------------------------------------------------------------------------------------------------|
| I   | Limitless replicative potential often results in the accumulation of chromosomal mutations in many cancer cells. |
| II  | Cancer cells could overproduce signal molecules so that they become self-sufficient in growth signals.           |
| III | Angiogenesis is the result of expression of oncogenes in a cell line that produces blood vessels.                |
| IV  | Loss-of-function mutations in tumour suppressor genes contribute to tissue invasion and metastasis.              |

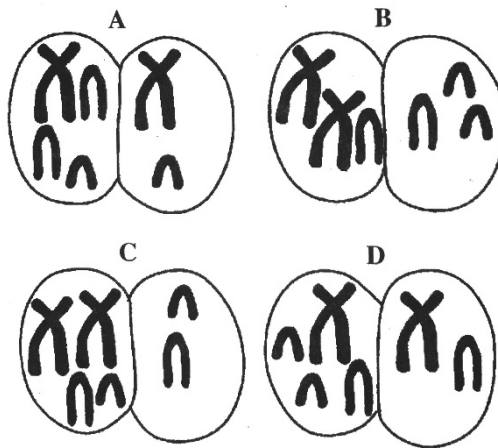
- A I and IV only
- B II and III only
- C I, II and IV only
- D II, III and IV only



10 The diagram shows three of the 23 pairs of chromosomes found in a human cell.



Which diagram shows an example of non-disjunction in the formation of an egg, that could lead to the formation of a Down's syndrome zygote?



11 A toxic chemical causes malfunction of the centrioles in animal cells.

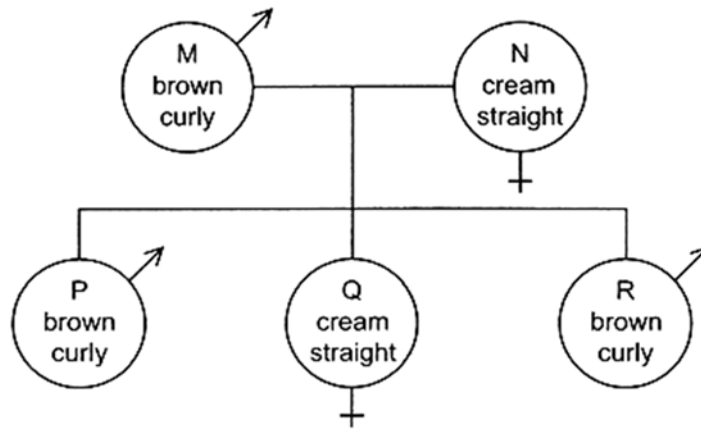
Which process in mitosis is likely to be directly affected by the chemical?

- 1 migration of the chromosomes
- 2 replication of centromeres
- 3 chromosome shortening
- 4 alignment of chromosomes

- A** 1 and 4 only  
**B** 2 and 4 only  
**C** 1, 3 and 4  
**D** 2, 3 and 4

- 12** Assume that in goats a pair of alleles is responsible for the inheritance of hair colour and that another pair controls hair texture. These pairs are located on different autosomes. The allele for brown hair (B) is dominant to the allele for cream hair (b), and the allele for curly hair (C) is dominant to the allele for straight hair (c).

The diagram shows a cross between two goats.



If R is mated with a female goat of the same genotype as M, what are the chances of the first offspring being a male with cream coloured, straight hair?

- A 0  
 B 1 in 32  
 C 1 in 16  
 D 1 in 4
- 13** A cross between a round-leafed, tall plant and round-leafed, dwarf plant produced the following offspring:

|                                |                |
|--------------------------------|----------------|
| 121 round-leafed, tall plants  | R – round leaf |
| 121 round-leafed, dwarf plants | r – oval leaf  |
| 42 oval-leafed, tall plants    | T – tall       |
| 37 oval-leafed, dwarf plants   | t – dwarf      |

What were the genotypes of the parents?

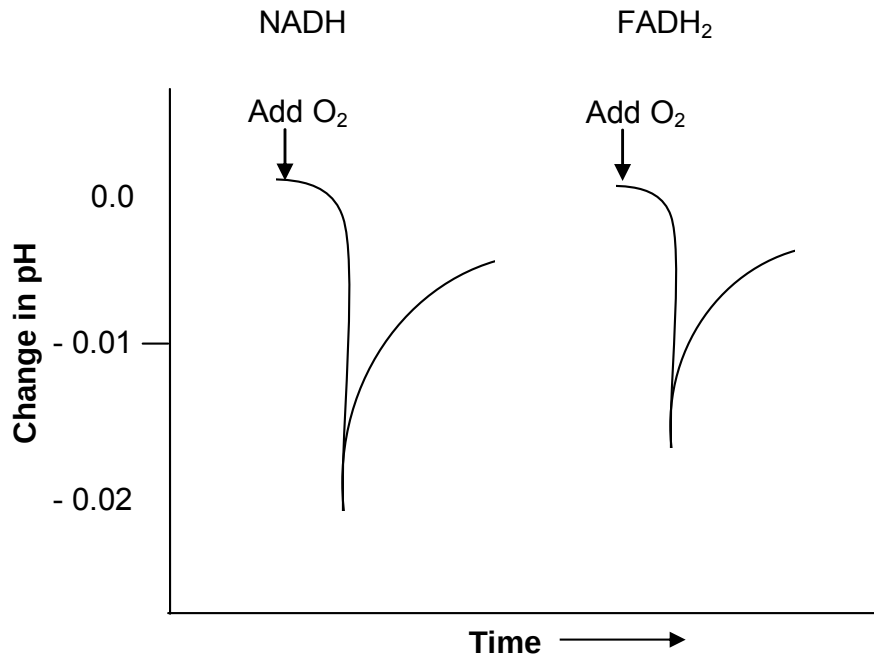
- A  $RrTt \times Rrtt$   
 B  $RrTt \times RRtt$   
 C  $RrTT \times Rrtt$   
 D  $RrTT \times RRtt$

- 14** Black, chestnut and bay (reddish brown with black mane, tail, eartips and lower legs) coat colour in horses is determined by two genes. The dominant E allele codes for black and red pigment, while the recessive e allele codes only for red pigment. At a separate gene, the dominant A allele restricts any black pigment produced to the horse's mane, tail, eartips and lower legs, while the recessive allele allows any black pigment produced to show up throughout.

Two black horses were mated. Which of the following coat colours will definitely **not** show up in their offspring?

- A** Bay
- B** Black
- C** Chestnut
- D** All three colours are possible

- 15 Isolated mitochondria were incubated with NADH in one experiment and an equal amount of FADH<sub>2</sub> in another experiment. The mitochondria were initially deprived of oxygen. The pH of the intermembrane space was then monitored as a known quantity of oxygen was added. The results are shown in the graph.

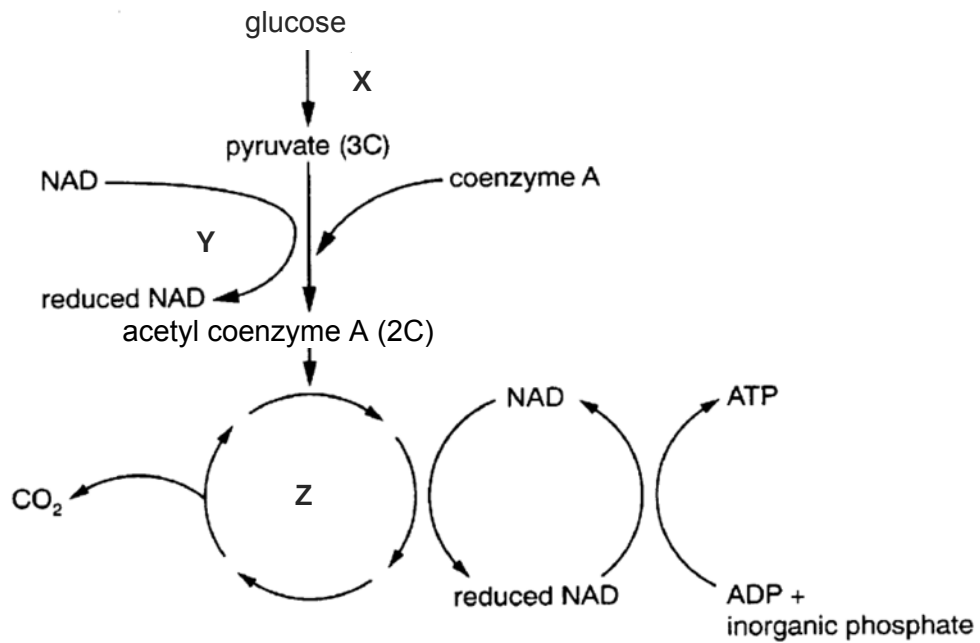


Which of the following can be concluded based on the results?

|     |                                                                                                                                                        |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| I   | Upon the addition of oxygen, glycolysis and subsequently, link reaction, Krebs cycle and oxidative phosphorylation occurred.                           |
| II  | Electron transfer was initiated by the addition of oxygen.                                                                                             |
| III | The pH drop was greater with NADH than with FADH <sub>2</sub> , which is consistent with the greater ATP yield that accompanies the oxidation of NADH. |
| IV  | The rapid decline in pH indicates that protons were pumped into the intermembrane space when oxygen was available.                                     |

- A I only  
 B II and IV only  
 C II, III and IV only  
 D All of the above

16 The flow chart shows a series of reactions occurring in an animal cell.



Which of the following statements correctly describes the flow chart?

- A Reaction X, which occurs in the cytosol, is an anabolic reaction.
- B Reaction Y involves the process of substrate-level phosphorylation, whereby pyruvate is first converted to a compound called acetyl coenzyme A.
- C Reaction Y occurs in the cytoplasm whereas reaction Z occurs in the mitochondria.
- D Reaction Z is a catabolic pathway which occurs twice for every glucose molecule to be completely oxidised.

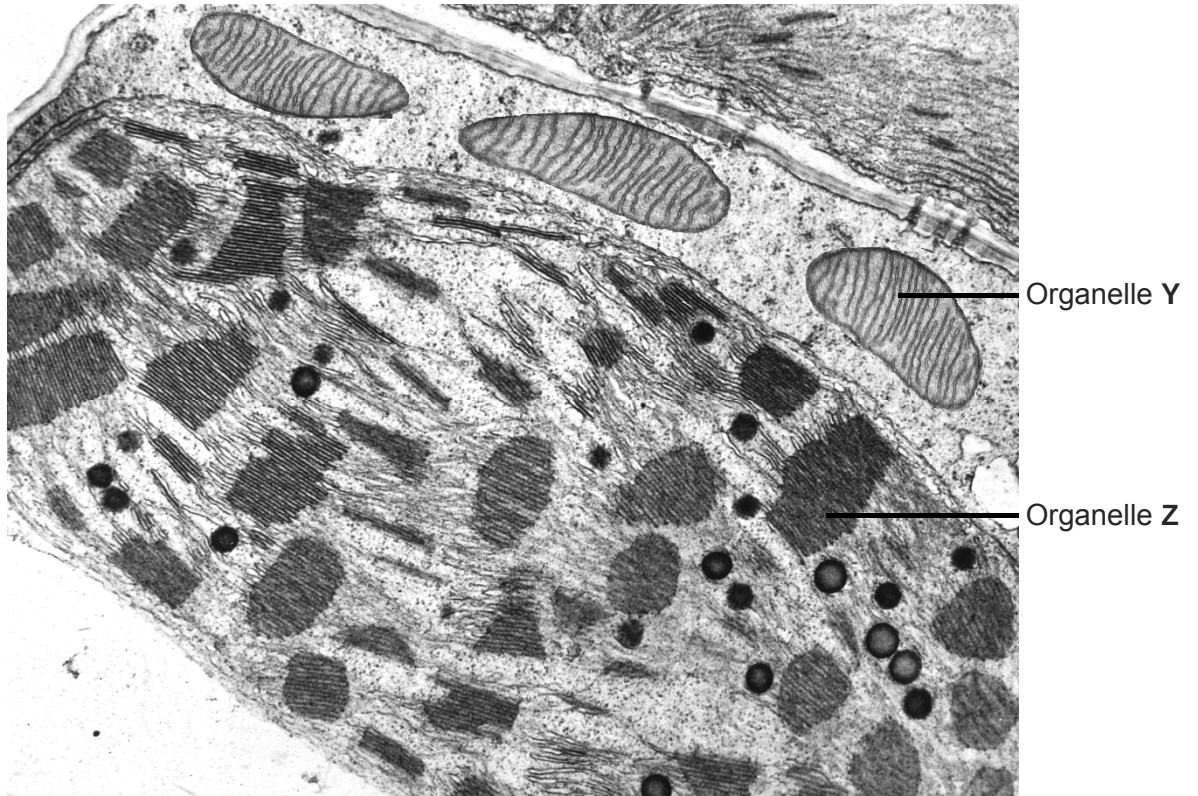
- 17 In a laboratory, a plant with variegated leaves (containing orange and green pigments) was supplied with radioactive carbon dioxide,  $^{14}\text{CO}_2$ . The plant was kept in the dark for 12 hours and then illuminated for the next 12 hours. A leaf from the plant was obtained and its level of radioactivity was measured both in the absence and presence of light. The results are shown in the table.

|                          | Level of radioactivity in leaf (Arbitrary units) |                      |
|--------------------------|--------------------------------------------------|----------------------|
|                          | Orange region of leaf                            | Green region of leaf |
| <b>Absence of light</b>  | 225                                              | 225                  |
| <b>Presence of light</b> | 410                                              | 9271                 |

Which of the following could be the most likely explanation for the level of radioactivity found in the orange region of the leaf in the presence of light?

- A Some photosynthesis occurs in the orange region but due to the absence of chlorophyll in that region, the rate of photosynthesis is low.
- B Photosynthesis occurs in the orange region but no storage of starch occurs.
- C Photosynthetic products diffuse into the orange region.
- D Radioactive  $^{14}\text{CO}_2$  diffuses into the orange region and accumulates in that region.

- 18 A new species of plant was recently discovered in the Amazon forest. The electron micrograph shows two organelles Y and Z in a leaf mesophyll cell of the plant.

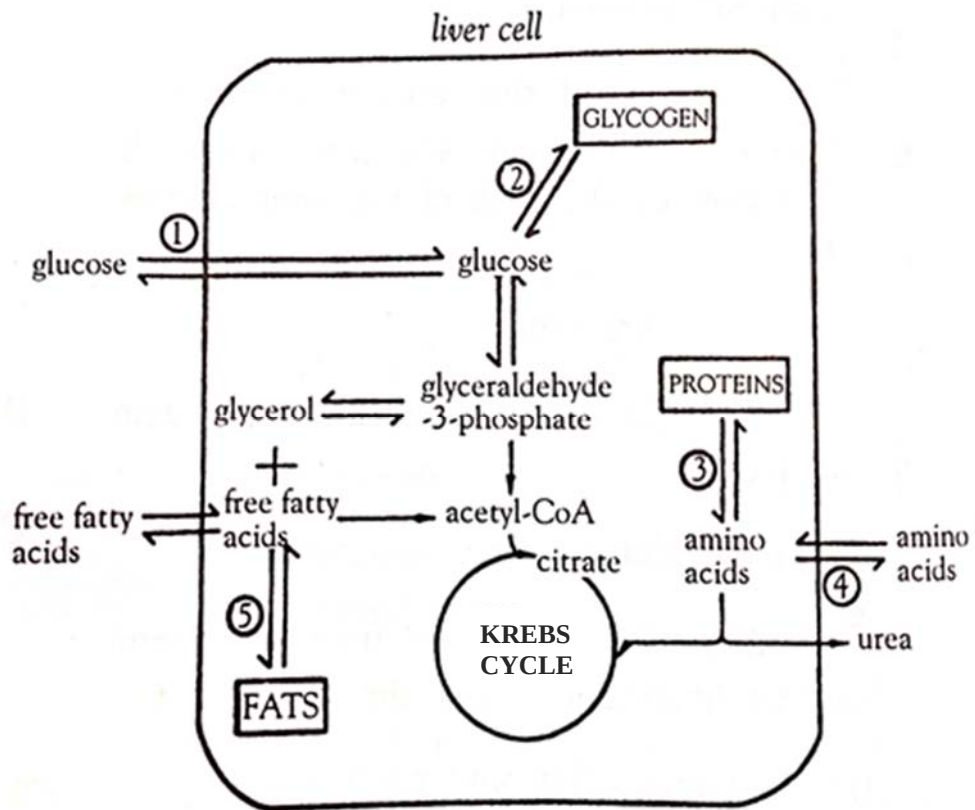


Which of the following statements are **not** true about organelles Y and Z?

|     |                                                                                                         |
|-----|---------------------------------------------------------------------------------------------------------|
| I   | Organelle Z utilises transporters to export ATP to organelle Y to drive cellular activities.            |
| II  | Oxygen released by organelle Z is used in organelle Y during Krebs cycle.                               |
| III | Phosphate ions in organelle Y is used for the production of ATP during Calvin cycle.                    |
| IV  | NADPH molecules produced in organelle Z are used in organelle Y for the production of triose phosphate. |

- A I and IV only  
 B II and III only  
 C II and IV only  
 D All of the above

- 19 The diagram shows some biochemical pathways in a liver cell. Some of the points where hormones affect the pathways are labelled 1 to 5.

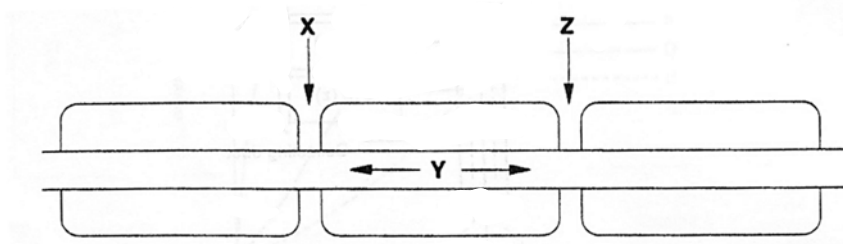


At which numbered points would the hormone insulin accelerate the pathway in the directions indicated?

- A 1, 2 and 3
- B 1, 2 and 5
- C 1, 3 and 4
- D 3, 4 and 5

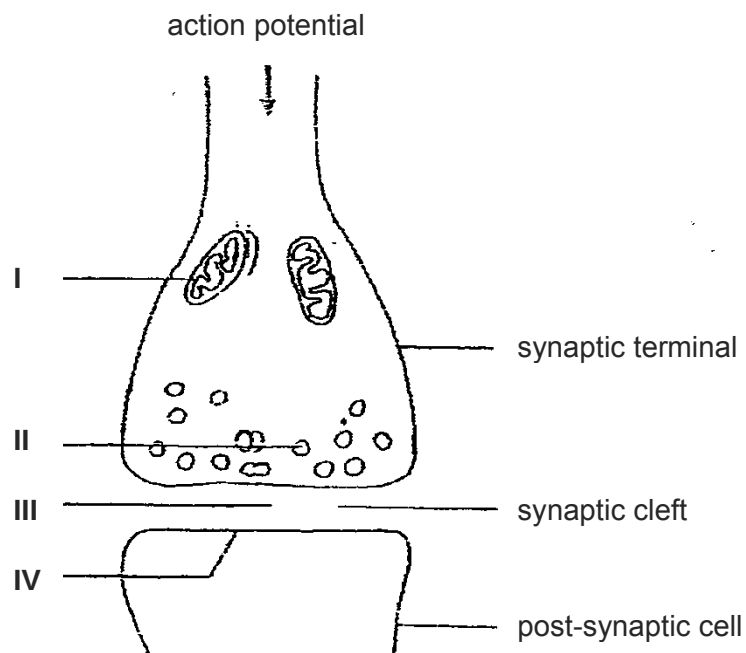


20 The diagram below shows a myelinated axon



How does the myelin sheath increase the speed of impulse transmission along the axon?

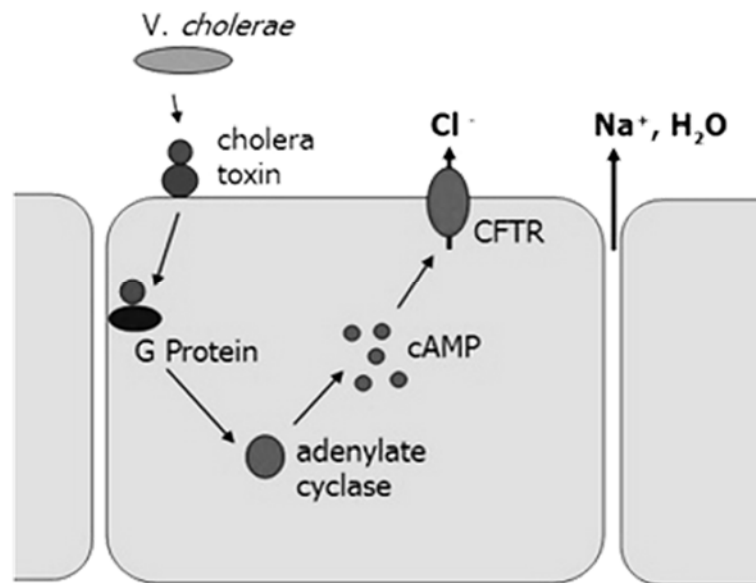
- A It ensures that the ions are kept close to the axon membrane in region Y.
  - B It insulates the axon, so increasing the potential at regions X and Z.
  - C It restricts the change in potential difference to regions X and Z.
  - D It promotes a change in potential difference in region Y.
- 21 Snake venom contains a neurotoxin that affects synaptic transmission. The toxin can cause paralysis and death.



Which of the following is **NOT** a possible way for the toxin to act?

- A The toxin reduces enzyme activity in structure I.
- B The toxin diffuses into structure II and binds to its contents.
- C The toxin binds with the contents of structure II after they are released into site III.
- D The toxin lodges itself in between the phospholipids found on site IV.

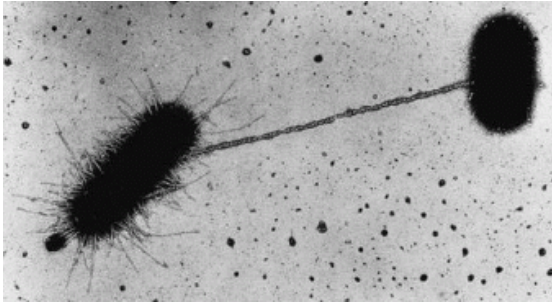
- 22** Cholera is a disease caused by infection of the intestine with the bacterium *Vibrio cholera*. This disease is characterised by profuse diarrhoea, leading to excessive loss of fluids and dehydration. Cholera toxin binds to receptor, resulting in the activation of G protein involved in regulating salt and water secretion.



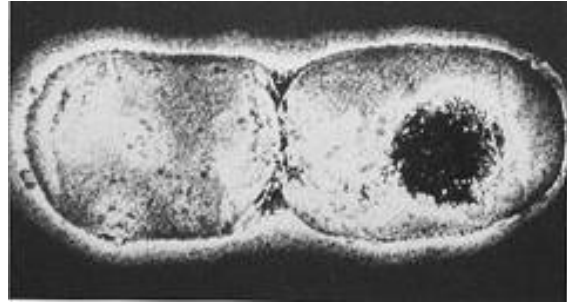
The toxin of *Vibrio cholera* causes profuse diarrhoea because

- A** cytosolic concentration of ions is decreased, making the cells hypotonic to the lumen of intestine
  - B** phosphodiesterase is permanently activated
  - C** G protein is modified such that it is unable to hydrolyse GTP to GDP
  - D** cystic fibrosis transmembrane conductance regulator (CFTR) Cl<sup>-</sup> channel is permanently activated due to binding of cAMP
- 23** What is an example of a step that amplifies the signal during its transduction in a cell?
- A** the action of adenylyl cyclase in converting ATP to ADP
  - B** the activation of protein kinase A by cAMP
  - C** the binding of a steroid hormone to its intracellular receptor
  - D** the phosphorylation of many mitogen-activated protein (MAP) kinase by an activated MAP kinase kinase

- 24** The photomicrographs below show two different processes occurring in two different species of bacteria.



Process 1



Process 2

Which of the following statements is/are true of both processes?

- (i) For both processes, only bacteria with genes that code for cytoplasmic bridge are involved.
- (ii) Process 1 requires direct contact between 2 different bacteria whereas process 2 can occur with 1 bacterium.
- (iii) Process 1 will result in an increase in the number of identical bacteria whereas process 2 will result in an increase in the number of different bacteria.
- (iv) Both processes involve DNA replication.

- A** (i) and (iii) only
- B** (ii) and (iv) only
- C** (i), (ii) and (iv) only
- D** (i), (ii), (iii) and (iv)

- 25** Which of the following statements could explain why a combination of different drugs rather than a single drug is being used to treat HIV patients?

- (i) HIV has a short generation span (eg. 2 days).
- (ii) In an AIDS patient, the HIV infection produces many new viruses per day (eg.  $10^{10}$  new viruses per day or more).
- (iii) HIV has an RNA genome which has a higher mutation rate than DNA as it is single stranded.
- (iv) The RNA genome allows the HIV to mutate rapidly to acquire the resistance to the drug being used.
- (v) Insertion of the HIV genome into the host cells will result in the host cells mutations that will confer resistance to the drug.

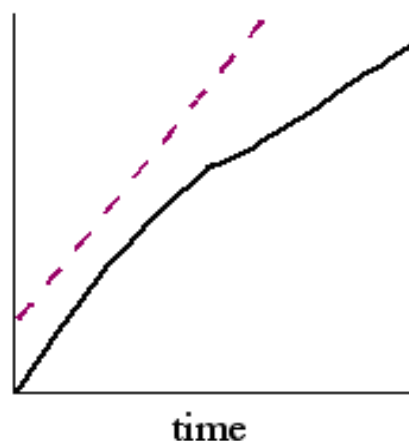
- A** (i) and (ii) only
- B** (iii) and (v) only
- C** (i), (ii) and (iii) only
- D** (iii), (iv) and (v) only

26 Which combination is true about bacteria?

|          | presence of methylated DNA | interaction of DNA with histone proteins | transcription initiation at promoter site |
|----------|----------------------------|------------------------------------------|-------------------------------------------|
| <b>A</b> | √                          | √                                        | √                                         |
| <b>B</b> | √                          | x                                        | √                                         |
| <b>C</b> | x                          | √                                        | √                                         |
| <b>D</b> | x                          | x                                        | x                                         |

27 The graph shows the bacterial growth and  $\beta$ -galactosidase production of mutant *Escherichia coli* cells (unable to produce *lacI* protein) with time.

- Bacterial growth is represented by solid line (—).
- $\beta$ -galactosidase production is represented by dashed line (- -).



Which of the following best explains the graph?

- A** high level of cAMP due to high concentration of glucose
- B** high level of cAMP due to low concentration of glucose
- C** low level of cAMP due to high concentration of glucose
- D** low level of cAMP due to low concentration of glucose

**28** Four different genes are regulated in different ways.

**Gene C:** regulatory gene whose product binds to an operator site

**Gene D:** product undergoes tissue-specific patterns of alternative splicing

**Gene E:** acetylation and deacetylation occurs to histones binding to the gene

**Gene F:** part of a group of structural gene controlled by the same regulatory sequence

Which combination correctly identifies which genes are prokaryotic and which are eukaryotic?

|          | <b>Prokaryotic</b> | <b>Eukaryotic</b> |
|----------|--------------------|-------------------|
| <b>A</b> | C and F            | D and E           |
| <b>B</b> | C and D            | E and F           |
| <b>C</b> | D and E            | C and F           |
| <b>D</b> | D and F            | C and E           |

**29** Which of the following is a feature of eukaryotic gene expression?

- A** Genes are organized in operons.
- B** Polycistronic mRNA is common.
- C** Transcription and translation are spatially separated.
- D** Translation initiation occurs with a molecule of formyl-methionine.

**30** How is translation controlled in eukaryotes?

- A** By differential removal of introns enabling a gene to code for more than one protein.
- B** By activation of the protein by folding or cleavage after it is formed.
- C** By the production of RNA from the non-coding strand of the DNA.
- D** By protein factors that bind to specific sequences in the mRNA.

**31** DNA methylation is known to silence genes because it prevents transcription factors from binding. Which of the following best explains this phenomenon?

- A** DNA methylation modifies the shape of the transcription factor.
- B** DNA methylation prevents dimerization of DNA binding proteins.
- C** DNA methylation modifies the shape of the DNA element where the transcription factor binds.
- D** DNA methylation causes acetylation of histone proteins which causes heterochromatin to be formed.

**32** Some statements concerning evolution are listed:

- 1** Offspring tend to resemble their parents.
- 2** Individuals in a sexually reproducing population are different.
- 3** The fossil record shows that many species have become extinct.
- 4** More offspring are produced than can possibly survive to sexual maturity.
- 5** Characteristics acquired during an organism's lifetime are passed to its offspring.

Which of these statements form the basis of Darwin's theory of evolution by natural selection?

- A** 1, 2 and 3
- B** 1, 2 and 4
- C** 1, 3 and 5
- D** 1, 4 and 5

**33** Which of the following pairs is *least* likely to represent homology?

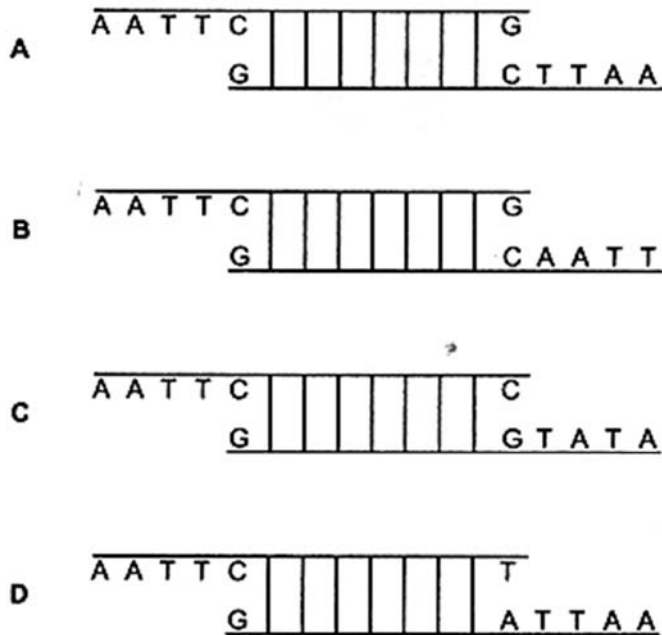
- A** the haemoglobin of a baboon and that of a gorilla
- B** the mitochondria of a plant and those of an animal
- C** the wings of a bird and those of an insect
- D** the wings of a bat and the arms of a human

**34** Which of the following statements does not correctly compare the neutral theory of molecular evolution and natural selection?

- A** Neutral theory of molecular evolution accounts for most of the differences that we observe at the phenotypic level as compared to natural selection.
- B** Neutral theory of molecular evolution accounts for most of the differences that we observe at the genotypic level as compared to natural selection.
- C** The rate of change in the nucleotide sequence brought about by neutral theory of molecular evolution occurs at a constant rate due to random chance events while the rate of change brought about by natural selection can be fast or slow depending on the strength of the selection pressure.
- D** Neutral theory of molecular evolution is largely responsible for the RFLP that we observe between species as compared to natural selection.

- 35 In genetic engineering, a restriction enzyme is used to cut plasmid DNA at a specific target site. The enzyme recognises a sequence of six bases and forms sticky ends.

Which diagram of such a cut section of DNA is correct?



- 36 The human genome project has identified and mapped the genes on human chromosomes. This is allowing scientists to identify specific, faulty genes which contribute to inherited conditions.

This is useful in many ways, for example

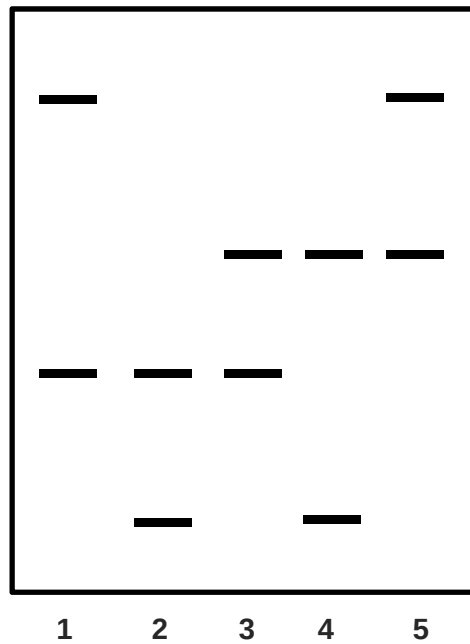
- 1 carriers of faulty genes can be advised about changes in lifestyle to minimise risks.
- 2 carriers of faulty genes can be identified and informed of their risk status.
- 3 diagnostic tests can be developed to identify carriers of faulty genes.
- 4 drugs can be developed to block the action of problem genes.
- 5 embryos can be screened to avoid the birth of affected children.
- 6 employers can take account of the genetic predisposition of employees.

Which two uses arise **directly** from the information provided by the project?

- A 1 and 2  
B 2 and 5  
C 3 and 4  
D 5 and 6



- 37 Results from the DNA fingerprint analysis of a single VNTR locus for a man and his 4 different children are shown in the autoradiograph.



Which lane contains the DNA of the father?

- A Lane 1
- B Lane 2
- C Lane 3
- D Lane 4

38 The common gene delivery system for in vivo gene therapy is

- I microinjection
- II liposome mediated gene transfer (lipofection)
- III electroporation
- IV adenoviruses

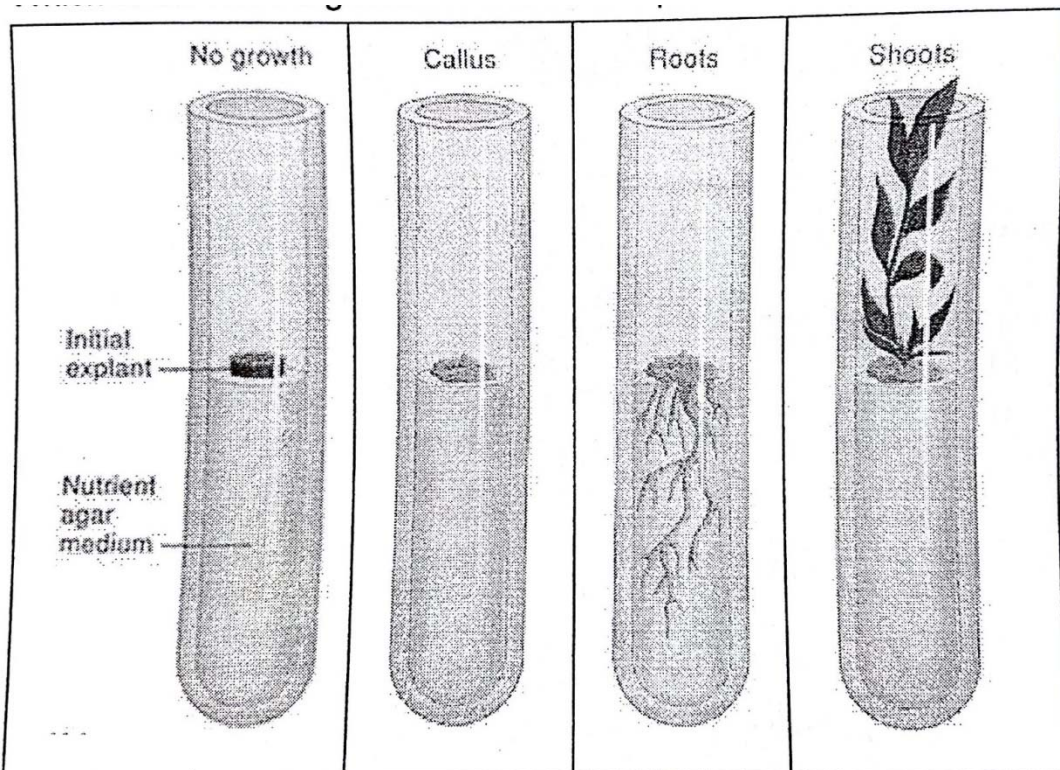
- A I only
- B I, II and III only
- C II and IV only
- D All of the above

39 A student wanted to investigate the response of an explant to various concentrations of IAA (auxin) and kinetin (cytokinin). He labelled and prepared the test tubes as follows:

| Test tubes     | 1    | 2    | 3    | 4    |
|----------------|------|------|------|------|
| IAA (mg/L)     | 0.02 | 0.00 | 1.00 | 2.00 |
| Kinetin (mg/L) | 1.00 | 0.20 | 0.20 | 0.02 |

After preparation, he handled the test tubes while his gloves were still wet with ethanol. Hence all labels were wiped off. After 2 weeks, his results were shown below.

Which of the following observations correspond to the correct test tube?



|   |        |        |        |        |
|---|--------|--------|--------|--------|
| A | Tube 3 | Tube 2 | Tube 1 | Tube 4 |
| B | Tube 2 | Tube 3 | Tube 4 | Tube 1 |
| C | Tube 1 | Tube 4 | Tube 3 | Tube 2 |
| D | Tube 4 | Tube 1 | Tube 2 | Tube 3 |

- 40** Which of the following statements best support the view that genetically modified crops could help resolve world food shortages?
- I** Genetic engineering enables production of drought resistant crops more quickly than selective breeding.
  - II** Genetically modified crops are produced by adding single genes.
  - III** Genetically modified crops can cross-fertilise with non-modified related plants.
  - IV** Genetically modified crops can be adapted to their environment when crossed with local varieties of the crop.
- A** I only
  - B** II and III only
  - C** II and IV only
  - D** I and IV only